

# Boran Aybak Kilic

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## Education

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|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>University of Illinois Urbana-Champaign</b><br><i>Exchange Student in Electrical &amp; Computer Engineering</i> | <b>2026 – Present</b><br><i>IL, USA</i>           |
| <b>Bogazici University</b><br><i>Double Major in Physics (4.0 GPA)</i>                                             | <b>2023 – Present</b><br><i>Istanbul, Turkiye</i> |
| <b>Bogazici University</b><br><i>B.Sc. in Electrical and Electronics Engineering (3.99 GPA)</i>                    | <b>2022 – Present</b><br><i>Istanbul, Turkiye</i> |

## Experience

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|------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Visiting Research Fellow</b><br><i>Harvard University, Physics Department, Lukin Group</i>                    | <i>June 2026- Sept 2026</i>  |
| <b>Research Assistant</b><br><i>University of Cambridge, Internet of Everything Group</i>                        | <i>Oct 2025 – Present</i>    |
| <b>Research Intern</b><br><i>Max Planck Institute for Human Cognitive and Brain Sciences, Neurophysics Dept.</i> | <i>June 2025 – Sept 2025</i> |
| <b>Summer Intern + Part-time Researcher</b><br><i>TÜBİTAK, Quantum Technologies Department</i>                   | <i>July 2024 – Jan 2026</i>  |
| <b>Undergraduate Research Assistant</b><br><i>Koc University, NWCL</i>                                           | <i>May 2024 – June 2025</i>  |
| <b>Deep Learning Apprentice</b><br><i>Neuromatch Academy</i>                                                     | <i>June 2024 – July 2024</i> |
| <b>Undergraduate Research Intern</b><br><i>Bogazici University, VAVLAB</i>                                       | <i>Feb 2024 – Present</i>    |

## Publications & Presentations

- **B. A. Kilic** and O. B. Akan, "Multi Ratio Shift Keying (MRSK) Modulation for Molecular Communication," *IEEE Transactions on Communications*, vol. 73, no. 10, Oct. 2025.
- Yüksel Dal D, **Kilic BA** and Gamgam G (2026) Local brain connectome parameters across the spectrum of clinical cognitive decline. *Front. Neurosci.* 20:1840382. doi: 10.3389/fnins.2026.1840382
- **B. A. Kilic** and O. B. Akan, "Neural-Inspired Multi-Agent Molecular Communication Networks for Collective Intelligence," *MOLCOM26* [abstract].
- **B. A. Kilic**, K. Pine, N. Weiskopf, "Synthetic Image and Multi-Coil K-space Data Generation From Multi-Parameter Maps," *ISMRM 2026*.
- Ö. Güngör, **B. A. Kilic**, et al., "Quantum Magnetic Anomaly Detection Using Nitrogen-Vacancy Centers Diamond," *KOBIT 2025* [Poster Presentation].

## Honors & Awards

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|-------------------------------------------------------------|---------------------|
| TUBITAK 2247-C Scholar                                      | Sept 2025 – Present |
| TUBITAK 2224-D Grant Recipient                              | June 2025           |
| TUBITAK 2205 Scholar                                        | 2022 – Present      |
| Ranked Top 1000 in National University Placement Exam (YKS) | 2022                |
| National Physics Olympiad, Bronze Medal (TÜBİTAK)           | 2019 & 2020         |

## Skills

- **Machine Learning:** GNNs, Transformers, CNNs, Transfer Learning, Attention-based Message Passing.
- **Scientific Expertise:** quantum optical computing, Quantitative MRI Physics, CV-QKD, Neuroimaging Pipelines
- **Modeling:** Stochastic Particle Simulations, Graph Theory, Information Theory, Monte Carlo methods.
- **Programming:** Python (PyTorch, NumPy, SciPy), MATLAB, C/C++, LaTeX, Git.
- **Languages:** Turkish (Native), English (Fluent), German (Intermediate).

## References

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**Prof. Burak Acar** (Bogazici University)  
*Principal Investigator, VAVLAB*

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**Prof. Özgür Barış Akan** (Cambridge University)  
*Principal Investigator, NWCL*

[oba21@cam.ac.uk](mailto:oba21@cam.ac.uk)

**Dr. Kerrin Pine** (MPI CBS)  
*Scientific Researcher, Neurophysics Dept.*

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